

## BLAJ, Romania

WMO: 152090

Lat: 46.1783N

Lon: 23.9353E

Elev: 338

StdP: 97.33

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -13.8      | -10.9 | -16.6                         | 0.9 | -12.5 | -13.5 | 1.2 | -9.9 | 16.2                  | -0.1 | 15.6 | 3.0  | 2.5                   | 0    | 0.511 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.6                   | 31.4            | 20.7 | 29.9 | 20.0 | 28.4 | 19.5 | 21.9                | 29.0 | 21.0 | 27.9 | 20.2 | 26.6 | 2.8                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.7                            | 15.0 | 24.5 | 18.8 | 14.1 | 23.2 | 18.1 | 13.6 | 22.4 | 65.3          | 29.1 | 62.2 | 28.1 | 59.5 | 26.6 | 24.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 15.8          | 10.5 | 9.4 | -18.2                      | 34.0 | 3.3                | 2.1 | -20.6                                              | 35.5 | -22.5      | 36.7 | -24.4      | 37.9 | -26.8      | 39.4 |    |    |
| (5)               |      |     | -18.3                      | 23.5 | 3.3                | 0.8 | -20.7                                              | 24.1 | -22.7      | 24.5 | -24.5      | 25.0 | -26.9      | 25.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |         | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|---------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |         | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg   | 10.4        | -1.9  | 1.6   | 5.4   | 11.1  | 15.8  | 19.5  | 20.7  | 21.0  | 16.4  | 10.7  | 4.8   | -0.3  |  |  |
|                                                                        | DBStd   | 8.92        | 4.59  | 4.83  | 4.28  | 4.23  | 3.25  | 2.97  | 2.64  | 2.81  | 4.10  | 3.99  | 4.63  | 4.62  |  |  |
|                                                                        | HDD10.0 | 1318        | 370   | 235   | 149   | 38    | 2     | 0     | 0     | 2     | 41    | 164   | 318   |       |  |  |
|                                                                        | HDD18.3 | 3151        | 628   | 468   | 399   | 218   | 91    | 21    | 11    | 9     | 85    | 238   | 405   | 577   |  |  |
|                                                                        | CDD10.0 | 1477        | 0     | 1     | 7     | 69    | 180   | 286   | 330   | 341   | 192   | 62    | 9     | 0     |  |  |
|                                                                        | CDD18.3 | 269         | 0     | 0     | 0     | 1     | 11    | 58    | 82    | 91    | 26    | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3 | 2528        | 0     | 0     | 1     | 30    | 141   | 498   | 709   | 863   | 275   | 11    | 0     | 0     |  |  |
|                                                                        | CDH26.7 | 753         | 0     | 0     | 0     | 1     | 17    | 127   | 213   | 314   | 79    | 1     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg   | 2.8         | 2.5   | 3.2   | 3.5   | 3.6   | 3.0   | 2.5   | 2.4   | 2.1   | 2.8   | 2.3   | 2.5   | 2.7   |  |  |
| (15) Precipitation                                                     | PrecAvg | 562         | 22    | 18    | 26    | 50    | 67    | 89    | 82    | 51    | 55    | 44    | 30    | 32    |  |  |
|                                                                        | PrecMax | 778         | 46    | 50    | 56    | 113   | 117   | 182   | 154   | 109   | 159   | 95    | 68    | 74    |  |  |
|                                                                        | PrecMin | 360         | 4     | 4     | 6     | 5     | 28    | 36    | 19    | 5     | 2     | 3     | 1     | 3     |  |  |
|                                                                        | PrecStd | 121         | 12    | 11    | 13    | 23    | 27    | 39    | 34    | 25    | 33    | 25    | 15    | 18    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%    | DB          | 9.0   | 16.3  | 21.2  | 26.2  | 28.6  | 31.9  | 32.5  | 34.3  | 32.1  | 24.9  | 18.9  | 11.4  |  |  |
|                                                                        |         | MCWB        | 5.9   | 9.0   | 11.2  | 15.2  | 17.6  | 21.4  | 21.5  | 21.6  | 20.1  | 16.3  | 12.6  | 7.6   |  |  |
|                                                                        | 2%      | DB          | 6.8   | 12.9  | 17.5  | 23.8  | 26.6  | 29.9  | 30.7  | 31.7  | 28.8  | 22.2  | 16.1  | 8.7   |  |  |
|                                                                        |         | MCWB        | 4.9   | 8.0   | 9.8   | 13.9  | 17.0  | 20.6  | 20.9  | 20.4  | 18.9  | 14.9  | 11.2  | 6.6   |  |  |
|                                                                        | 5%      | DB          | 5.0   | 10.7  | 14.8  | 21.3  | 24.8  | 28.0  | 29.1  | 30.0  | 26.4  | 20.1  | 13.9  | 6.8   |  |  |
|                                                                        |         | MCWB        | 3.4   | 6.9   | 8.5   | 12.7  | 16.3  | 19.7  | 20.2  | 19.8  | 17.3  | 13.7  | 10.2  | 5.2   |  |  |
|                                                                        | 10%     | DB          | 3.3   | 8.5   | 12.4  | 18.9  | 23.0  | 26.3  | 27.5  | 28.2  | 24.2  | 17.8  | 11.8  | 5.2   |  |  |
|                                                                        |         | MCWB        | 2.2   | 5.5   | 7.1   | 11.8  | 15.5  | 19.0  | 19.5  | 19.4  | 16.7  | 12.9  | 8.9   | 4.2   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%    | WB          | 6.6   | 10.3  | 12.0  | 16.1  | 18.7  | 23.0  | 23.0  | 22.8  | 20.8  | 17.2  | 13.1  | 8.8   |  |  |
|                                                                        |         | MCDB        | 8.3   | 15.0  | 19.9  | 24.0  | 26.0  | 29.8  | 29.4  | 31.6  | 30.1  | 22.5  | 17.4  | 10.1  |  |  |
|                                                                        | 2%      | WB          | 5.0   | 8.3   | 10.1  | 14.6  | 17.6  | 21.4  | 21.9  | 21.5  | 19.3  | 15.7  | 11.8  | 6.5   |  |  |
|                                                                        |         | MCDB        | 6.7   | 12.1  | 16.4  | 22.6  | 24.5  | 27.7  | 28.4  | 29.5  | 27.0  | 20.3  | 15.1  | 8.3   |  |  |
|                                                                        | 5%      | WB          | 3.6   | 7.1   | 8.8   | 13.3  | 16.9  | 20.4  | 21.0  | 20.7  | 18.3  | 14.5  | 10.6  | 5.5   |  |  |
|                                                                        |         | MCDB        | 4.8   | 10.3  | 14.0  | 19.8  | 23.4  | 26.5  | 27.4  | 28.1  | 24.6  | 18.9  | 13.6  | 6.8   |  |  |
|                                                                        | 10%     | WB          | 2.3   | 5.7   | 7.6   | 12.2  | 16.0  | 19.6  | 20.2  | 19.9  | 17.1  | 13.3  | 9.2   | 4.3   |  |  |
|                                                                        |         | MCDB        | 3.2   | 8.4   | 11.6  | 17.6  | 21.8  | 25.1  | 25.8  | 26.5  | 22.4  | 17.2  | 11.6  | 5.3   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB   | MDBR        | 5.8   | 8.5   | 10.3  | 11.5  | 11.1  | 11.3  | 11.8  | 12.6  | 11.5  | 10.8  | 7.7   | 5.1   |  |  |
|                                                                        |         | MCDBR       | 7.4   | 12.4  | 14.8  | 16.2  | 14.7  | 14.2  | 14.8  | 15.8  | 15.7  | 15.8  | 11.8  | 7.5   |  |  |
|                                                                        |         | MCWBR       | 5.5   | 7.7   | 8.2   | 7.4   | 6.2   | 5.8   | 5.7   | 5.4   | 6.2   | 8.3   | 7.1   | 5.6   |  |  |
|                                                                        |         | MCWBR       | 7.1   | 11.6  | 13.4  | 14.3  | 13.6  | 13.0  | 13.4  | 14.5  | 14.4  | 13.8  | 10.7  | 7.0   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub    | taub        | 0.312 | 0.340 | 0.381 | 0.435 | 0.419 | 0.415 | 0.434 | 0.442 | 0.400 | 0.365 | 0.339 | 0.313 |  |  |
|                                                                        |         | taud        | 2.351 | 2.299 | 2.241 | 2.118 | 2.226 | 2.270 | 2.224 | 2.197 | 2.285 | 2.371 | 2.390 | 2.372 |  |  |
|                                                                        |         | Ebn at Noon | 777   | 824   | 837   | 821   | 849   | 853   | 830   | 805   | 809   | 786   | 738   | 731   |  |  |
|                                                                        |         | Edh at Noon | 75    | 96    | 118   | 146   | 136   | 131   | 136   | 134   | 113   | 89    | 73    | 67    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg  | RadAvg      | 1.28  | 2.10  | 3.21  | 4.33  | 5.26  | 5.78  | 5.71  | 5.24  | 3.75  | 2.58  | 1.49  | 1.01  |  |  |
|                                                                        |         | RadStd      | 0.16  | 0.26  | 0.42  | 0.45  | 0.49  | 0.51  | 0.49  | 0.49  | 0.41  | 0.31  | 0.19  | 0.13  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (79 neighbors) | +0.74 | N/A     | N/A    | +0.99   | N/A   | N/A   | -149        | -215    | +92     | +23     |

Nomenclature: See separate page